

Assignment Choice 2 – Trip Fare Estimation

Description:

You are a newly recruited data science team member for Gett, New York City branch, and the company is undergoing a challenging period since it only recently entered the ride-sharing market in NYC and is still “learning” the market and how it functions in order to tailor its business operations effectively to the people and userbase in NYC.

In your first week at work, your manager called you in for an urgent meeting with the Marketplace team head. Once you went into the meeting room with them and sat down, your manager said “Hope you took your DSMA course during your Bachelor’s degree at BGU, you’re going to need to use everything that you learned to get us through this difficult period” and handed you a 3 page document that lists the business question which the team urgently needs investigated. It reads as follows:

What is the trip’s total fare amount estimate we must project to the app’s user? If we overcharge, Gett’s passengers won’t book the ride. If we undercharge, our drivers will protest.

- a. **Input** – PULocationID, DOLocationID, trip_distance, Time of day, Day of week
- b. **Output** (target variable) – *total_fare_amount* (*total_amount* – *tip_amount*)
- c. **The challenge:** unlike the trip_duration prediction problem, *total_amount* prediction is highly sensitive to all the values that are added up for the fare. This problem will require a significant amount of feature engineering to get high accuracy and low error rates.
 - i. **Extensive Feature Engineering** - You will need to think how to model all the different values that are a part of the *total_fare_amount* before the trip has occurred.
 1. Go through the data dictionary for the problem to understand how *total_amount* is calculated.
 2. Understand that none of the money related features from the dataset are provided as part of the input since the business usecase that you are building here is trip fare estimate – which means you inform the users of the Gett Application how much their trip is likely to cost before they press the “book taxi” button. (you’re only providing an estimate)
 3. Therefore, all the features like whether there is a *congestion_surcharge*, etc. as part of the *total_amount* will have to be calculated using other data that is available to you (mentioned in **a. Input** above) at inference time as part of your feature engineering.
 4. You may also need to use external data that is not present within the dataset (by explore the NYC TLC website in order to understand how surcharges are calculated, etc.)
 5. Approximate those values and use the values in the dataset to see if your logic at approximating those values are working well. Try searching for fare rules in the NYC TLC website and build the feature engineering pipe accordingly (domain – based features)

- ii. **Handle outliers** – you need to be thorough with your data cleaning and validation before you begin working with the data. (check for errors like negative fares)
- d. **Build a simple Streamlit UI** for the purpose of showcasing how your final model(s) predict the total_fare_amount of a trip – have a few input boxes for each of the inputs and a button which says “book taxi”. The UI should display a map that shows a point to point trip with an overlay displaying how much the customer will probably be charged for their trip.
- e. **Data use constraints:** You may use the data over 2024 and 2025 as you wish, but reserve 2026 data specifically for evaluating and demonstrating how your model performs. (Do not include 2026 data in your training set.)

After staring at this document with beads of sweat forming on your forehead, you turn to leave the meeting room, when the Marketplace team head laughs and remarks “Thankfully, the municipality of New York has open sourced its historical data on traditional taxi rides in the city. There should be more than enough data in those data dumps that we can analyse to answer these questions.” You run quickly to your laptop and run a search online to find these datasets and discover the following:

Dataset:

NYC – TLC (year of 2024 and 2025, and JAN + FEB of 2026) – can be downloaded from here - <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

Use only the “**Yellow Taxi Trip Records**” datasets from **2024, 2025** and the **first two months of 2026**.

Data Dictionary:

Can be found here -

https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_yellow.pdf

And the Taxi Zone Lookup Table can be found here:

https://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv

Use of AI:

Permitted, but please put in a little effort to understand the code that is generated (and to check if it is correct / useful for what you wish you achieve) before you use it in the project.

Evaluation Criteria:

1. **Completeness (20%)** – the extent to which the machine learning pipeline was built for answering this business question.
2. **Effectiveness of EDA (20%)** – How well you explored the data to find the initial insights in the data that allowed you to work effectively with the data.
3. **Innovation (20%)** – be creative with your feature engineering and model development
4. **Modularity of the Code (20%)** – Make the codebase that you build for this assignment modular and readable. One of the goals of this course is to highlight how modularity of the codebase makes it much simpler to maintain and work with (especially when working in teams).

5. **Performance (10%)** – how well your model(s) work over the test set (you score 10/10 here if you beat the lecturer's model).
6. **Use of Experiment Tracking (10%)** – In the course, we use Weights & Biases (wandb) for the purpose of experiment tracking. I strongly encourage using experiment tracking when running your experiments (for the purpose of hyperparameter tuning using sweeps, for running evaluations of your models before and after you make changes to your features in order to see what works, and to also work with model and data versioning (as we will see in Lecture 4))

Assignment Submission format:

You are expected to submit 3 files:

1. **A python notebook** that you developed for performing the EDA that the lecturer can run over the training data (2024 + 2025 combined)
2. **A zip folder containing the entire codebase** you developed for the machine learning pipeline which the lecturer can execute that will run everything needed for the pipeline to work from start to finish. **Training data** – 2024 + 2025, **Test Data** – Jan + Feb of 2026
(include a text file containing instructions to execute the code if you feel it's needed)
3. **A PDF report** summarizing the following:
 - a. Your key findings from the EDA connecting it to the business problem that you are trying to solve for. (e.g., Airport rides seem to have a bump in total_amount since the fare there is fixed.)
 - b. A short explanation of the final selected set of features that you engineered
 - c. A short explanation of how you performed model selection and hyperparameter tuning
 - d. A short explanation of the model improvement loops that you did (error analysis of your model + drift detection and mitigation)
 - e. A link to your experiment tracking page on wandb (within the same wandb team that you have joined for the course. You can open a new project for yourself.)